



DesignWare SDRAM and Static Memory Controller Release Notes

DW_memctl

Copyright Notice and Proprietary Information

Copyright © 2012 Synopsys, Inc. All rights reserved. This software and documentation contain confidential and proprietary information that is the property of Synopsys, Inc. The software and documentation are furnished under a license agreement and may be used or copied only in accordance with the terms of the license agreement. No part of the software and documentation may be reproduced, transmitted, or translated, in any form or by any means, electronic, mechanical, manual, optical, or otherwise, without prior written permission of Synopsys, Inc., or as expressly provided by the license agreement.

Destination Control Statement

All technical data contained in this publication is subject to the export control laws of the United States of America. Disclosure to nationals of other countries contrary to United States law is prohibited. It is the reader's responsibility to determine the applicable regulations and to comply with them.

Disclaimer

SYNOPSYS, INC., AND ITS LICENSORS MAKE NO WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, WITH REGARD TO THIS MATERIAL, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE.

Registered Trademarks (®)

Synopsys, AEON, AMPS, ARC, Astro, Behavior Extracting Synthesis Technology, Cadabra, CATS, Certify, CHIPit, CoMET, Confirma, CODE V, Design Compiler, DesignSphere, DesignWare, Eclipse, Formality, Galaxy Custom Designer, Global Synthesis, HAPS, HapsTrak, HDL Analyst, HSIM, HSPICE, Identify, Leda, LightTools, MAST, MaVeric, METeor, ModelTools, NanoSim, NOVeA, OpenVera, ORA, PathMill, Physical Compiler, PrimeTime, SCOPE, SiVL, SNUG, SolvNet, Sonic Focus, STAR Memory System, SVP Café, Syndicated, Synplicity, the Synplicity logo, Synplify, Synplify Pro, Synthesis Constraints Optimization Environment, TetraMAX, UMRBus, VCS, Vera, and YieldExplorer are registered trademarks of Synopsys, Inc.

Trademarks (™)

AFGen, Apollo, ASAP, Astro-Rail, Astro-Xtalk, Aurora, AvanWaves, BEST, Columbia, Columbia-CE, Cosmos, CosmosLE, CosmosScope, CRITIC, CustomExplorer, CustomSim, Custom WaveView, DC Expert, DC Professional, DC Ultra, Design Analyzer, Design Vision, DesignerHDL, DesignPower, DFTMAX, Direct Silicon Access, Discovery, Encore, EPIC, Galaxy, HANEX, HDL Compiler, Hercules, Hierarchical Optimization Technology, High-performance ASIC Prototyping System, HSIMplus, i-Virtual Stepper, IC Compiler, IICE, in-Sync, iN-Tandem, Intelli, Jupiter, Jupiter-DP, JupiterXT, JupiterXT-ASIC, Liberty, Libra-Passport, Library Compiler, Macro-PLUS, Magellan, Mars, Mars-Rail, Mars-Xtalk, Milkyway, ModelSource, Module Compiler, MultiPoint, ORAengineering, Physical Analyst, Planet, Planet-PL, Platform Architect, Polaris, Power Compiler, Processor Designer, Raphael, RippledMixer, Saturn, Scirocco, Scirocco-i, SiWare, SPW, Star-RCXT, Star-SimXT, StarRC, Symphony Model System Compiler, System Designer, System Studio, Taurus, TotalRecall, TSUPREM-4, VCSi, VHDL Compiler, VMC, and Worksheet Buffer are trademarks of Synopsys, Inc.

Service Marks (SM)

MAP-in and TAP-in are service marks of Synopsys, Inc.

Third Party Trademark Acknowledgements

SystemC is a trademark of the Open SystemC Initiative and is used under license.

ARM and AMBA are registered trademarks of ARM Limited.

Saber is a registered trademark of SabreMark Limited Partnership and is used under license.

Entrust is a registered trademark of Entrust Inc. in the United States and in certain other countries. In Canada, Entrust is a trademark or registered trademark of Entrust Technologies Limited. Used by Entrust.net Inc. under license.

All other product or company names may be trademarks of their respective owners.

Synopsys, Inc.
700 E. Middlefield Road
Mountain View, CA 94043

www.synopsys.com

Contents

Preface	5
Chapter 1	
DW_memctl Release Notes	7
1.0.1 DW_memctl New Features and Changes	7
1.0.2 DW_memctl Known Problems and Workarounds	11
1.1 STARs on the Web	12
1.2 Global Synthesizable IP Issues	12
Appendix A	
Pre-October 2007 DW_memctl STARs	13
A.1 STARs on the Web	13
A.2 DW_memctl Archives	13

Preface

This manual describes release issues regarding DesignWare SDRAM and Static Memory Controller (DW_memctl) and contains information of interest to anyone using the DW_memctl.

Release Notes Organization

The chapters of this databook are organized as follows:

- Chapter 1, “[DW_memctl Release Notes](#)” lists the features, issues, and past releases related to the DW_memctl.
- Chapter A, “[Pre-October 2007 DW_memctl STARS](#)” contains archived STAR tables for the DW_memctl.

Related Documents

To see a complete listing of documentation available for the DesignWare synthesizable and verification components for the DW_memctl, refer to the [Guide to Documentation for DesignWare SDRAM and Static Memory Controller](#).

To see installation instructions for the DW_memctl, refer to the [DesignWare SDRAM and Static Memory Controller Installation Guide](#).

Web Resources

- DesignWare IP product information: <http://www.designware.com>
- Your custom DesignWare IP page: <http://www.mydesignware.com>
- Documentation through SolvNet: <http://solvnet.synopsys.com> (Synopsys password required)
- Synopsys Common Licensing (SCL): <http://www.synopsys.com/keys>

Customer Support

To obtain support for your product:

- First, prepare the following debug information, if applicable:
 - For environment setup problems or failures with configuration, simulation, or synthesis that occur within coreConsultant or coreAssembler, use the following menu entry:
File > Build Debug Tar-file

Check all the boxes in the dialog box that apply to your issue. This menu entry gathers all the Synopsys product data needed to begin debugging an issue and writes it to the file `<core tool startup directory>/debug.tar.gz`.

- For simulation issues outside of coreConsultant or coreAssembler:
 - Create a waveforms file (such as VPD or VCD)
 - Identify the hierarchy path to the DesignWare instance
 - Identify the timestamp of any signals or locations in the waveforms that are not understood
- Then, contact Support Center, with a description of your question and supplying the above information, using one of the following methods:
 - For fastest response, use the SolvNet website. If you fill in your information as explained below, your issue is automatically routed to a support engineer who is experienced with your product. The **Sub Product** entry is critical for correct routing.

Go to <http://solvnet.synopsys.com/EnterACall> and click on the link to enter a call. Provide the requested information, including:

- **Product:** DesignWare Library IP
- **Sub Product:** Memory
- **Tool Version:** *<product version number>*
- **Problem Type:**
- **Priority:**
- **Title:** DW_memctl
- **Description:** For simulation issues, include the timestamp of any signals or locations in waveforms that are not understood

After creating the case, attach any debug files you created in the previous step.

- Or, send an e-mail message to support_center@synopsys.com (your email will be queued and then, on a first-come, first-served basis, manually routed to the correct support engineer):
 - Include the Product name, Sub Product name, and Tool Version number in your e-mail (as identified above) so it can be routed correctly.
 - For simulation issues, include the timestamp of any signals or locations in waveforms that are not understood
 - Attach any debug files you created in the previous step.
- Or, telephone your local support center:
 - North America:
Call 1-800-245-8005 from 7 AM to 5:30 PM Pacific time, Monday through Friday.
 - All other countries:
<http://www.synopsys.com/Support/GlobalSupportCenters>

DW_memctl Release Notes

This chapter presents the latest release information about the DW_memctl.

Refer to the [coreTools Release Notes](#) for a list of known problems and limitations in coreConsultant and coreAssembler.

The following sections list the new features and changes, and the fixed problems and enhancements, for the DW_memctl component.

For DW_memctl-specific STARs, refer to STARs on the Web (SotW):

https://www.synopsys.com/dw/doc.php/iip/DW_memctl/latest/doc/dmctl_db.pdf

1.0.1 DW_memctl New Features and Changes

The following was new or changed in the 2.79c version of the DW_memctl:

- Decoupled DW_memctl from dw_iip_amba image

The following was new or changed in the 2.79b version of the DW_memctl:

- RTL changes:
 - None
- Documentation and/or coreTools changes:
 - Uses coreTools version 2010.09-SP2

The following was new or changed in the 2.79a version of the DW_memctl:

- RTL changes:
 - Added three registers for bus turn around timing parameter (T_bta) for static memories; t_bta_0_cnt, t_bta_1_cnt, and t_bta_2_cnt
 - Added the correct driving of s_we_n_wpipe
 - Fixed problem with floating nets sd_in_dpd_mode and s_rp_n
 - Fixed situations in which unused nets occur in various configurations
 - Generated RAL file no longer includes registers not present in the selected configuration

- ❑ DWMM installation activity can now be bypassed when memory models are not required
- ❑ Replaced reuse-pragmas for coreTools-packaged Perl with generic location to `#!/usr/bin/perl`
- Documentation and/or coreTools changes:
 - ❑ Updated system graphic in Figure 1-1

The following was new or changed in the 2.78b version of the DW_memctl:

- RTL changes:
 - ❑ Added C headers to component package
 - ❑ Removed obsolete registers from C and verilog headers and power-on reset register value file
 - ❑ Added code for missing component parameter, type, and version registers
- Documentation and/or coreTools changes:
 - ❑ Removed code for SYNC_FLASH_OPCODE, SFCONR, SFCTLR, and SFTMGR in example in Programming chapter
 - ❑ Added material for component type, version, and parameter registers

The following is new or has changed in the 2.78a version of the DW_memctl:

- RTL changes:
 - ❑ None
- Documentation and/or coreTools changes:
 - ❑ Uses coreTools version 2010.03 or later
 - ❑ Corrected names of include files and vcs command used for simulation in databook

The following was new or changed in the 2.76a version of the DW_memctl:

- RTL changes:
 - ❑ DW_memctl no longer uses x86_linux versions of the memory models
 - ❑ Simulation now uses correct register base address
- Documentation and/or coreTools changes:
 - ❑ Updated description of TWP and TWS
 - ❑ Updated description of low_sync_device
 - ❑ Corrected register description of LOW_FREQ_DEV_SET n
 - ❑ Clarification that only one type of SDRAM is supported per configuration
 - ❑ Updated function mux_addr in “Verilog RTL for EBI Logic” section
 - ❑ Updated Table 7 footnote
 - ❑ Uses coreTools version 2008.06-SP2-2

The following was new or changed in the 2.75b version of the DW_memctl:

- RTL changes:
 - None
- Documentation and/or coreTools changes:
 - Uses coreTools version 2007.06-SP4

The following was new or changed in the 2.75a version of the DW_memctl:

- Multiple bugs fixed (see SotW): http://www.synopsys.com/dw/star.php?c=DW_memctl
- Multiple documentation updates

The following was new or changed in the 2.74b version of the DW_memctl:

- Using upgraded versions of supported tools; refer to the *DesignWare SDRAM and Static Memory Controller Installation Guide*.

The following was new or changed in the 2.74a version of DW_memctl

- Changed references from DesignWare Connect to coreAssembler across the DW_memctl documentation set, where appropriate.
- Updated the automated DesignWare Memory Model software/model download and installation process.
- Mobile DDR-SDRAM DW Memory Models are now available to aid in DW_memctl configuration and for use in “Verify Component” simulations.
- “Verify Component” simulations on 64-bit operating systems now automatically execute 32-bit simulations on all supported simulators. 32-bit Vera and memory models are also required.
- Support for coreTools 2006.03-SP3

The following was new or changed in the 2.73a version of the DW_memctl:

- The databook is now split into a user manual and a databook in order to better organize the information into user material and reference material. A flow diagram was also added to the first chapter of the user manual in order to illustrate the usage flow of the component.

The following was new or changed in the v2.72a version of the DW_memctl.

- A new flow tutorial based on DesignWare Connect now comprises Chapter 2, “[Building and Verifying a Component or Subsystem](#),” of the *DesignWare SDRAM and Static Memory Controller Databook*.
- The DW_memctl Driver Kit is available, which allows you to easily program the DW_memctl and integrate it into your higher-level application. For information about the DW_memctl Driver Kit, refer to the *DesignWare DW_memctl Driver Kit User Guide*.
- The DW_memctl now supports Mobile DDR-SDRAM.
- The DW_memctl no longer needs a Vera license for regressions.
- The DW_memctl can now be used in the coreTools 5.x environment.
- The DW_memctl now supports the DC-FPGA environment.

- The size of the DesignWare Synthesizable Components image has been reduced to about 60 MB by removing the DesignWare Memory Model TSP and the QuickStart examples designs; these are now available through separate downloads. For more information, refer to the [DesignWare SDRAM and Static Memory Controller Installation Guide](#).
- The DesignWare Synthesizable Components image is now self-extracting.

For a list of all the features of the DW_memctl, refer to the “Features” section of the [DesignWare SDRAM and Static Memory Controller Databook](#).

The following was new or changed in the 2.71a version of the DW_memctl.

- Source code for this component is available on a per-project basis as a DesignWare Core. Please contact your local sales office for the details. For source licensing information, refer to “Licenses” in the *DesignWare SDRAM and Static Memory Controller Installation Guide*.
- DesignWare Connect – Enables you to construct, modify, and simulate any single- or multi-layer system in about an hour.

The following changes occurred in the 2.70b version of the DW_memctl.

- The following SDR-SDRAMs from Elpida Memory, Inc. and Nanya Technology Corp. are not supported by the DW_memctl because they do not support burst termination:
 - hm5264805f nt5sv16m16at
 - hm5264405f nt5sv16m16atl
 - hm5264165f nt5sv16m4dt
 - hm5259805b nt5sv16m8ct
 - hm5259405b nt5sv32m4ct
 - hm5259165b nt5sv32m8at
 - hm5257805b nt5sv4m16dt
 - hm5257405b nt5sv64m4at
 - hm5257165b nt5sv8m16ct
 - hm5225805b nt5sv8m8dt
 - hm5225405b
 - hm5225165b
- Cadence Verilog-XL v4.0 and NC-Verilog v4.0 simulations are not supported on the HP platform.

The following changes occurred in the 2.60a version of the DW_memctl.

- The power-on bus width of the Static memory mapped to “SET0” registers is controlled by the sm_data_width_set0[2:0] port. In earlier versions of DW_memctl, it was controlled by the SM_DATA_WIDTH_SET0 parameter. This enhancement enables users to configure the FLASH/ROM boot data width through strap pins, instead of needing to fix the data width at synthesis time.

1.0.2 DW_memctl Known Problems and Workarounds

The following are known issues in this release of the DW_memctl:

- STAR 9000083421 – In one scenario, t_{rc} is programmed to 64 cycles. When a read burst on the AHB side comes in with $hsize = 64$ bits, the read on the static memory side violates t_{rc} for the fifth data element. In this situation, the data width for the static memory is 16 bits, so the first beat of AHB read data takes four beats to the Static memory. The first four beats work correctly and there are no violations of t_{rc} . However, after the fourth beat, the Static memory address changes to the next address, but it takes two clock cycles for the chip select and oe to de-assert. Thus a memory connected to this bus, finishes the fourth beat access and then starts on a new access to the next address, but is then interrupted two clock cycles later.

When $hsize = 32$ bits, everything runs properly; that is, after the second Static memory beat, the chip select and oe de-assert at the correct time.

- STAR 9000083992 – Problem: Bus turn-around timing parameter (T_{bta}) for Static memories.

In one scenario, the DW_memctl has two Static memories connected at chip select1 and chip select2 - CS[1] and CS[2]. The device at CS[1] has a T_{bta} value of 7, while the device at CS[2] has a T_{bta} value of 1. If a write access to CS[1] is followed immediately by a write access to CS[2] and again an immediate read to CS[1], the last read violates T_{bta} (7 clocks in this example)

This problem exists because the DW_memctl does not keep track of the T_{bta} value for the previous chip select. The T_{bta} counter looks only at the current chip select's T_{bta} value.

Going from CS[1] to CS[2], the t_{bta} counter is loaded with the t_{bta} value for CS[2], which in this case is a value of 1. Thus, when it goes back to CS[1] immediately with a read, the DW_memctl asserts CS[4] before the T_{bta} of "7" for CS[1] completes.

Workaround: Increase the t_{bta} value of the second chip select, CS[2]. However, this affects the timing of the DW_memctl. In this example, set the CS[2] T_{bta} to 3; specify either at configuration or during runtime, by writing into the SMTMGR_SET0.

- STAR 9000041326 – AHB INCR write/read transfer termination inserts additional memory cycles – There are one or two extra clock cycles inserted on the memory bus at the end of an INCR AHB WR/RD full burst; this is due to the registered termination that is implemented for better timing. Extra write cycles have write enable and byte select signals masked, which does not cause data corruption. However, there is a potential performance issue with small INCR bursts. Extra read cycles at the end of INCR are not propagated to the bus interface.
- STAR 9000035185 - DW_memctl can inherit the wrong timing parameters from certain memory models.

For the mt48lc16m16a SDRAM memory model and 75MHz clock frequency specified, the -trp value that the coreConsultant GUI "inherits" from the DWMM is three clocks. From the memory datasheet, the Trp value should be 20 ns. Since the clock frequency is 75MHz, the Trp value should be two clocks for a 20ns window.

Workaround: Review and verify the automatically set configuration parameters in the GUI.

- DW_memctl is not optimized for partial burst transfers on AHB – All partial bursts are internally converted to single transfers, and burst access protocol to a memory – especially DRAM – is not exploited, which incurs performance loss. An example of this is when a burst transfer with HSIZE of 16 bits is performed on a 32-bit wide AHB configuration.

- The following SDR-SDRAMs from Elpida Memory, Inc. and Nanya Technology Corp. are not supported by the DW_memctl because they do not support burst termination:

hm5264805f	nt5sv16m16at
hm5264405f	nt5sv16m16atl
hm5264165f	nt5sv16m4dt
hm5259805b	nt5sv16m8ct
hm5259405b	nt5sv32m4ct
hm5259165b	nt5sv32m8at
hm5257805b	nt5sv4m16dt
hm5257405b	nt5sv64m4at
hm5257165b	nt5sv8m16ct
hm5225805b	nt5sv8m8dt
hm5225405b	
hm5225165b	

1.1 STARS on the Web

You can view problem reports for components used in this release, including problems identified after product release, by accessing the STAR report on the Web. Note that you must have a SolvNet ID in order to view STAR reports. You can access STAR reports for any verification or synthesizable IP component through the IP Directory:

<http://www.synopsys.com/dw/ipsearch.php>

1.2 Global Synthesizable IP Issues

For a list of release-related issues for the AMBA 2/AMBA 3 AXI Synthesizable IP components, refer to:

http://www.synopsys.com/dw/doc.php/doc/amba/latest/dw_amba_rn.pdf

A

Pre-October 2007 DW_memctl STARs

This appendix contains archived STAR tables for DW_memctl.

A.1 STARs on the Web

You can view problem reports for components used in this release, including problems identified after product release, by accessing the STAR report on the Web. Note that you must have a SolvNet ID in order to view STAR reports. You can access STAR reports for any verification or synthesizable IP component through the IP Directory:

<http://www.synopsys.com/dw/ipsearch.php>

For DW_memctl-specific STARs, refer to:

http://www.synopsys.com/dw/star.php?c=DW_memctl

A.2 DW_memctl Archives

The following tables describe the DW_memctl STARs that were fixed in each of the versions prior to October 2007.

Table A-1 STARs Fixed in DW_memctl Version 2.74a

STAR ID	Type	Description
STS0184246	Enhancement	Add simulation setup check for overlap in register address space and memory device address space.
9000062474	Enhancement	Name clash error when bus_naming_style is %s_%d.
9000068040	Enhancement	Handle non-automatic generated block size in simulation setup.
9000074472	Enhancement	Simulation IWNF warnings for s_rd_ready and s_sda_in signals.
9000113824	Enhancement	Add models for Mobile DDR-SDRAM during Choose Memory Models activity.

Table A-1 STARs Fixed in DW_memctl Version 2.74a

STAR ID	Type	Description
9000115927	Bug	For designs with two or more DW_memctl instances that are configured differently for “SDRAM Controller Type” (DYNAMIC_RAM_TYPE parameter) or “Add pipeline to address decoder outputs” (PIPE_ADDRESS_DECODER parameter), there can be parameter contamination from one core to the other. Define statements from the first DW_memctl instance incorrectly affect other DW_memctl instances, resulting in incorrect netlists or simulations.
9000116926	Bug	Update Databook and GUI to better document DWMM software and model download.
9000117025	Bug	Installing the downloaded DWMM models within coreConsultant causes an error when working with the DW_memctl for the first time.
9000117116	Bug	coreConsultant and coreAssembler attempt to use Netscape even when Mozilla is the chosen browser.
9000117119	Bug	When a workspace is reopened in either coreAssembler or coreConsultant, the previous installation of LMC_HOME is not detected.
9000123366	Bug	Update Databook to clarify OPEN BANK parameter and register field.
9000129197	Bug	Cannot use 32-bit DW memory models on AMD64 platform.
9000132271	Bug	Simulations fail on 64-bit Solaris with 64-bit DW memory models.
9000132756	Bug	Failure on deep power-down mode simulation.
9000132757	Bug	Update Databook/Application Note documentation of deep power-down mode.
9000132758	Bug	Update Databook for t_init requirements for deep power-down mode.
9000136736	Enhancement	Ports of DW_memctl_miu_sdc are unconnected at instantiation, causing simulation warnings.
9000138103	Bug	When configured as SDRAM controller with only one chip select, simulations can fail if SDRAM Base Address changed from default.
9000170048	Bug	Update Databook to explicitly state that sm_ready requires T_AS>0.

Table A-2 STARs Fixed in DW_memctl Version 2.73a

STAR ID	Type	Description
9000028117	Bug	coreConsultant produces an error when the “Enable SDRAM and Static Memory Data bus Sharing” parameter is set instead of a message.
9000029940	Bug	DW_memctl documentation should clearly mention that it supports NOR-Flash memories.
9000062750	Bug	Component simulations fail in coreAssembler (DesignWare Connect) but pass in coreConsultant.

Table A-2 STARs Fixed in DW_memctl Version 2.73a (Continued)

STAR ID	Type	Description
9000081728	Bug	DWMM installation fails within DesignWare Connect (coreAssembler).
9000081729	Bug	DesignWare Connect (coreAssembler) downloads an old sl_admin so that amd64 does not work for the DWMM.
9000083409	Bug	In some scenarios, the DW_memctl violates t_rp timing parameter. The violation occurs between a precharge command and an auto-refresh command. This situation can arise when the user shifts from one chip select to another. The DW_memctl does not load the T_rp counter at that time. In some corner cases, the t_ref period ends at the wrong time: the clock idle cycle that occurs between the close (precharge) of the previous bank and before the active command to the new chip select. This violates the t_rp timing parameter to the memory at the first chip select.
9000083446	Bug	Incorrect Specify Configuration activity and the ability to allow Controller Chip Selects cause simulation failures.
9000083449	Bug	Simulations do not start without \$SYNOPSYS being defined even when using VCS.
9000103449	Bug	An error occurs in coreConsultant and coreAssembler concerning the “head” command having obsoleted the “-#” option and requiring “n #”.
9000103916	Bug	Default value for T_WR in coreConsultant cannot be set to the maximum value of 4.

Table A-3 STARs Fixed in DW_memctl Version 2.72a

STAR ID	Type	Description
9000056318	Bug	A 64-bit value for the AHB Address width parameter (H_ADDR_WIDTH) is not supported as a configuration option of the design, although the manual mentions the possibility. In the GUI, H_ADDR_WIDTH is fixed to 32-bits. There is support for a 64-bit address width in RTL, but it has not been verified and there is no support to test it in a verification environment.
9000056774	Bug	The coreConsultant plugins cannot correctly read the value of PAGE_MODE_SETn for a given Static memory. This causes simulations to fail when using memories with page_mode operation AND when relying on the DW_memctl to read the correct value for page_mode from the memory model. For page mode memories, the T_PRC_SETn timing value is observed on the first page read access, and subsequent accesses to the same page are governed by T_RC_SETn. For non-page mode memories, the relevant timing parameter is T_RC_SETn; T_PRC_SETn is not read from the DWMMTable. The workaround is to manually override the PAGE_MODE_SETn value.

Table A-3 STARs Fixed in DW_memctl Version 2.72a

STAR ID	Type	Description
9000057587	Bug	If a user configures the DW_memctl as a Static memory controller and sets write_pipe, the simulation from within coreConsultant fails. This happens because the DW_memctl testbench has an error. Currently the chip selects for Static memories are incorrectly generated based on the `WRITE_PIPE` value. For correct operation, it should be derived from the `SM_WRITE_PIPE` value. WRITE_PIPE should be used for Dynamic memories, and SM_WRITE_PIPE for Static memories. Note that this is a bug in the coreConsultant verification environment, NOT an RTL bug and does not affect the functionality of the core.
9000007539	Bug	INCR4 read inserted BUSY. This caused the read data overflow to occur, and the HIU terminated the memory burst and tried to rebuild the burst. The HIU monitored the HTRANS to become non-BUSY and pushed the HADDR to the address FIFO to rebuild the memory burst. Unfortunately in this case, the INCR4 read was terminated at BUSY, so the read to the last data never occurred and the address the HIU captured belonged to the next transfer.
9000020778	Bug	During a read, the cycle time is basically 32/width times the programmed cycle time; that is, 4x or 2x the programmed time. This is also documented in some DW_memctl FAQ. This behavior is all right from a functional perspective, but impacts system performance with external 16-bit program FLASH.

Table A-4 STARs Fixed in DW_memctl Versions 2.70a and 2.70b

STAR ID	Type	Description
176940	Bug	The Intro Doc needs to be corrected. On page 2, the DW_memctl/doc/intro.pdf, dated April 1, 2003, reads: “Application Notes DesignWare DW_memctl Application Notes - Procedural information and concepts for using the DWPCI in designs.” “DWPCI” should be changed to “DW_memctl.”
181269	Bug	DDR write-data toggles even when no access made to memory. The DDR-SDRAM write-data output was changing every clock, even when no access to DW_memctl is made. This would cause more power consumption. Muxing the high and low write data to the DDR-SDRAM was not qualified with DW_memctl select, causing the unnecessary data toggle.

Table A-4 STARs Fixed in DW_memctl Versions 2.70a and 2.70b

STAR ID	Type	Description
184742	Bug	With EBI enabled, the DW_memctl begins initializing the memory device, even if ebi_gnt is not granted to the DW_memctl. This causes the initialization to fail. Workaround: Directly connect the s_addr[10] signal to that of the memory device by bypassing the EBI. You can connect the DW_memctl s_addr [10] directly to the device address bit 10. Thus the memory device address bus connection would look like: <pre>{ebi_addr[16:11], s_addr[10], ebi_addr[9:0]}</pre> However, the EBI bus would receive the s_addr from DW_memctl as {s_addr[16:11, 1'b0, s_addr[9:0]}.
184746	Bug	DW_memctl DDR data outputs toggle bug still exists. This STAR is the same as star 181269, which was reported as fixed in version 2.60A. However, this bug did not get fixed correctly in 2.60A and still exists. This new STAR reports this fact. The data outputs from the controller switch between the high and low halves of the write data value at the DDR rate, even when no write operation is in progress. This behavior can cause excessive power consumption.

Table A-5 STARs Fixed in DW_memctl Version 2.60a

STAR ID	Type	Description
181915	Enhancement	Static-Memory Bus Turnaround time (t_bta) enhancement. The DW_memctl provided 1-clock bus turn around time between read to read when the chip-select changes. With slower memories and high AHB clock speed this may not be sufficient. Hence the “t_bta” function has been enhanced to avoid bus-contention during back-to-back reads to two different chip-selects. The “t_bta” definition has been changed from: Static memory idle cycles between “read to write”, or “write to read” for memory data bus turn around time; values of 0-7 correspond to 0-7 idle clock cycles. to: Static memory idle cycles between “read to write,” or “write to read,” or “read to read when chip-select changes” for memory data bus turn-around time; values of 0—7 correspond to 0—7 idle clock cycles.

Table A-6 STARs Fixed in DW_memctl Version 2.50a

STAR ID	Type	Description
166788	Bug	Reads to SCORN and SFCORN registers do not return correct value. If you use SDRAM and set HARD_WIRE_SDRAM_PARAMETERS to 1, reads to SCONR (0x0) and SFCONR(0xB0) do not return correct value. This does not affect SDRAM or SyncFlash operation, but affects the Serial Presence Detect feature of SDRAM DIMMs, since SCONR is used for reading EEPROM data (bit 20). Workaround: Always set HARD_WIRE_SDRAM_PARAMETERS to 0.

Table A-6 STARs Fixed in DW_memctl Version 2.50a

STAR ID	Type	Description
173330	Bug	Standard coreConsultant configuration report not available. DW_memctl provides custom report associated with Specify Configuration activity, but does not generate standard coreConsultant configuration report.

Table A-7 STARs Fixed in DW_memctl Version 2.4a

STAR ID	Type	Description
161657	Bug	LMC_HOME variable is nullified if a workspace is closed and then reopened. When an user downloads memory models from the web through the coreConsultant GUI, the coreKit automatically sets LMC_HOME to the DWMM installation area. If the user closes the workspace and reopens it again, the LMC_HOME variable is not set again, which causes an "LMC_HOME" not set error and hence a simulation error. Workaround – Set the LMC_HOME variable in your environment (.cshrc) to the DWMM installation area. This has to be done before you invoke coreConsultant. You should point the LMC_HOME path to where the "models" directory is located in your DesignWare Memory Model installation path. If you are using the DWMM installed by coreConsultant, then you should point LMC_HOME to <i>install_dir/DW_memctl/2.3a/DWMM</i> .
164448	Bug	Problem with DW_memctl 2.3a and SRAM-only configuration. A problem exists when using DW_memctl 2.3A to configure the controller as an SRAM-only interface. Even though the user configures the data width to 128 bits for the AHB interface and the memory, the parameter that is written out for S_DATA_WIDTH is 32. Workaround – Manually edit the <i>src/DW_memctl_params.v</i> file and correct S_DATA_WIDTH to 128.
164449	Bug	With DW_memctl 2.3a, FC2 scripts have an error. A bug exists when using DW_memctl 2.3a and coreConsultant 3.2.3 to generate scripts for FC2. Two separate lines in the script are joined together as a single line in the script that is written out by coreConsultant. Workaround – Edit the <i>workspace/syn/fpga/scripts/DW_memctl.cscr</i> file and separate the following lines with a return: <pre> set_input_delay -group (RC,hclk) 15 \ DW_memctl/test_mode set_input_delay -group (RC,hclk_2x) 7 \ DW_memctl/test_mode </pre>

Table A-7 STARs Fixed in DW_memctl Version 2.4a

STAR ID	Type	Description
164451	Bug	The output bus of the DW_memctl is toggling even when the controller is not selected. The memory write-data output of the DW_memctl toggles even when the DW_memctl is not selected. (The registered version of the AHB "hwdata" is output to the memory write data bus). Although this is not a logic error, it could increase power consumption. Workaround – Currently there is no software workaround for this bug.
165543	Bug	DW_memctl hangs when EBI is enabled in SDR-SDRAM+SYncFlash (COMBO) mode. When DW_memctl is configured as an SDR-SDRAM and SyncFlash controller together (COMBO MODE), and the EBI (External Bus Interface) is enabled, the EBI grant signal is incorrectly assigned internally to 0. Under these circumstances, the DW_memctl hangs and waits for the EBI grant. This does not happen in any other DW_memctl configurations, even when using EBI mode. Workaround – Currently there is no workaround for this bug.

Table A-8 STARs Fixed in DW_memctl Version 2.3a

STAR ID	Type	Description
160250	Bug	Self-refresh does not work when SDRAM is put in self-refresh mode. Ideally, self-refresh should occur when all these signals are low: CAS, RAS and CKE (Clock Enable) and CS (chip select). This bug occurs during self-refresh because the CS pin does not go low at all, which means that the device never enters self-refresh mode. This situation can cause data corruption. This bug occurs in only versions 2.0 and later. Workaround - none at this time.
160347	Bug	DW_memctl does not end a burst transaction if only HSEL is de-asserted. This bug affects users only if they use only the Slave Select (HSEL) to force early termination. The DW_memctl does not recognize AHB early termination, which is generated by deactivating the Slave select (HSEL). In order for the DW_memctl to recognize the early termination, HTRANS must go IDLE or NONSEQ. Workaround - Early termination should be generated by adapting HTRANS, rather than de-activating the slave select (HSEL).

Table A-9 STARs Fixed in DW_memctl Version 2.2a

STAR ID	Type	Description
158610	Bug	SDR/Mobile-SDRAM incorrect t_{wr} during burst-termination The t_{wr} parameter definition was inconsistent or not defined during burst-write termination. Even though t_{wr} parameter was defined as last write data-in to precharge, it started from burst-termination (the last data-in is previous clock). This affected only SDR/Mobile-SDRAM (not DD-SDRAM) controllers in versions 2.0 and 2.1a (the previous version used precharge instead burst-termination). Workaround for v2.1a: Increase the t_{WR} parameter by one.

Table A-10 STARs Fixed in DW_memctl Version 2.1a

STAR ID	Type	Description
150574	Bug	The page_bank_sf test will not compile with Vera restrictive compile, since it has “posedge” constructs.
152099	Bug	A mask signal inside the DW_memctl address decoder logic was incorrectly sized to 11 bits instead of 16. Because of this, bits [15:11]

Table A-11 STARs Fixed in DW_memctl Version 2.0a

STAR ID	Type	Description
146807	Bug	A problem exists when the DW_memctl is configured to operate in big-endian mode. Currently, even if big-endian mode is configured, reads/writes to registers require data to be provided in little-endian mode.
146941	Bug	There is a problem in the remap address decoder logic. Currently chip select becomes active when: 1. AHB address matches Chip-Select Base address (or) 2. AHB address matches Remap Base address and “remap” pin is high The correct behavior should be: 1. AHB address matches Chip-Select base address and “remap” pin is low (or) 2. AHB address matches Remap Base address and “remap” pin is high
STAR ID	Type	Description
133958	Bug	Address-FIFO overflow when early termination during wrapping burst. In cases of wrapping bursts, if early termination happens after the first cycle followed by IDLE and another memory access to the memory controller, Address-FIFO may overflow.

STAR ID	Type	Description
134485	Bug	Error in Static memory FLASH/register writes if AHB:memory data ratio is not 1:1. When the AHB:Memory data width is not 1:1, during FLASH writes, each write causes more than 1 write (AHB/Memory writes). This results in incorrect FLASH programming. Also, when the AHB to SDRAM data width ratio is not 1:1, the Static memory register writes could corrupt the Static memory configuration registers for a clock.

Table A-12 STARs Fixed in DW_memctl Version 1.1

STAR ID	Type	Description
133589	Bug	In cases where wrapping bursts were terminated early before the wrapping took place, and was immediately followed by another burst to a different memory (different chip select), the first burst was not being properly terminated on the memory side.

